

SLIP – 15

Section I: Data Structures [15 Marks]

Q 1) Sort a random array of n integers (accept the value of n from user) in ascending order by using insertion sort algorithm.

Answer:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int a[100], n, i, j, key;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
    }
    return 0;

    for(i = 0; i < n; i++)
        a[i] = rand() % 100;

    printf("Original array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    for(i = 1; i < n; i++)
    {
        key = a[i];
        j = i - 1;

        while(j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j--;
        }
        a[j + 1] = key;
    }

    printf("\nSorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
}
```

```
    return 0;
}
```

Section II: Database Management Systems [15 Marks]

Q 2) Create the following tables (primary keys are underlined).

Game (gname, noofplayers, coachname, captain_name)

Player (pno, pname)

There exists a one-to-many relationship between Game and Player.

Constraint captain_name should be uppercase.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Game (
    gname VARCHAR(50) PRIMARY KEY,
    noofplayers INT,    coachname
    VARCHAR(50),    captain_name
    VARCHAR(50)
    CHECK(captain_name = UPPER(captain_name))
);
```

```
CREATE TABLE Player(
    pno INT PRIMARY KEY,
    pname VARCHAR(50),
    gname VARCHAR(50) REFERENCES Game(gname)
);
```

```
INSERT INTO Game VALUES
('hockey',11,'Singh','SANDEEP'),
('kho kho',9,'Patil','AMIT'),
('cricket',11,'Dhoni','VIRAT');
```

```
INSERT INTO Player VALUES
(1,'Ravi','hockey'),
(2,'Sunil','kho kho'),
(3,'Anil','cricket'),
(4,'Kiran','hockey');
```

Q 2(B)

B. Execute the following queries in PostgreSQL (any 3) [6 marks]

i. List the names of players playing “hockey”.

Answer:

```
SELECT pname FROM Player WHERE  
gname = 'hockey';
```

ii. Give the average number of players.

Answer:

```
SELECT AVG(noofplayers) FROM Game;
```

iii. Delete the records of players playing ‘kho kho’.

Answer:

```
DELETE FROM Player  
WHERE gname = 'kho kho';
```

iv. List names of players not playing “cricket”.

Q 2(C)

C. Write a query to display the game details with maximum number of players. [4 marks]

Answer:

```
SELECT * FROM Game  
WHERE noofplayers = (SELECT MAX(noofplayers) FROM Game);
```

Q 3) Viva [5 marks]